

PROFILE

I use high-fidelity simulation and interpretable computational models to study biological locomotion and collective dynamics. I am translating that foundation toward scientific machine learning, embodied intelligence, and swarm robotics.

CURRENT

Ph.D. Candidate, Mechanical Engineering

Johns Hopkins University

PLANNED DEFENSE

December 2026

CORE CAPABILITIES

- Immersed-boundary direct numerical simulation
- Physics-informed multi-agent systems
- Scientific machine learning and active learning
- HPC simulation campaigns and scientific visualization

SELECTED WORK

2026

JOURNAL OF FLUID MECHANICS

Scaling laws for caudal fin swimmers

DNS dataset spanning Reynolds numbers 5,000-100,000.

2025

JOURNAL OF FLUID MECHANICS

Hydrodynamically beneficial school configurations in carangiform swimmers

Simulation design, computational analysis, and manuscript preparation.

2025

PHYSICS OF FLUIDS - COVER ARTICLE

Effect of hydrodynamic wakes in dynamical models of large-scale fish schools

Wake-coupled model development, numerical experiments, and analysis.

2024

NEURIPS

ActSort: An active-learning accelerated cell sorting algorithm

Domain-informed features, data curation, benchmarking, and evaluation.